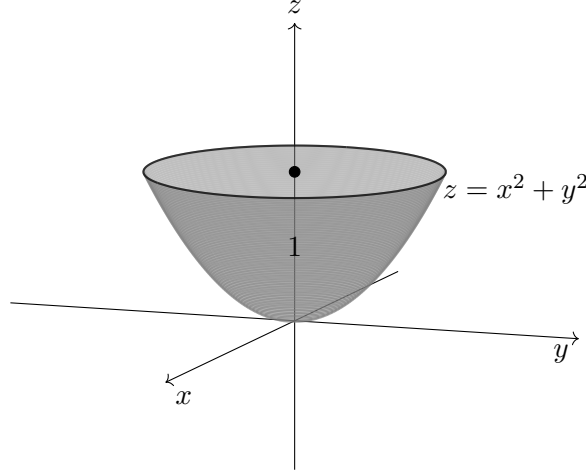


1. Calculate the volume this shape encloses, in all the three coordinate systems. The height of this shape is 1 unit along the z -axis. Comment on which system gives simpler calculations and why that is the case.



2. In spherical coordinates (r, θ, ϕ) (θ is the polar angle), a particle follows the path given by

$$r(t) = 1, \quad \theta(t) = \frac{\pi}{4} + 0.5t, \quad \phi(t) = 4t$$

Compute its velocity and acceleration vectors at $t = 0$

3. Consider the vector field in $\mathbb{R}^3$

$$\mathbf{V}(\mathbf{r}) = \frac{\hat{\mathbf{r}}}{r^2} = \frac{\mathbf{r}}{r^3}, \quad r = \|\mathbf{r}\|,$$

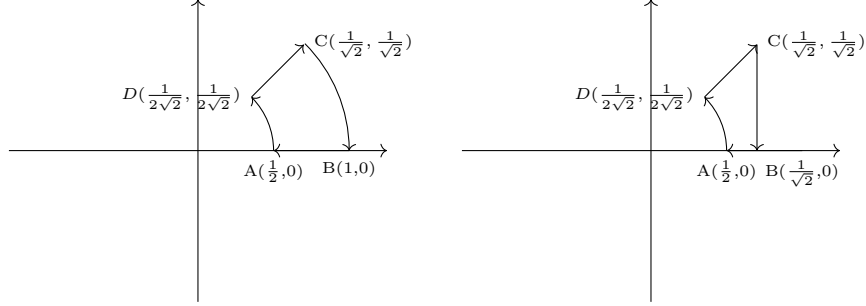
defined for all $\mathbf{r} \neq \mathbf{0}$. The field is singular at the origin and is often interpreted as the velocity field of an idealized point source (radial outflow) in an incompressible fluid.

- Show that** $\nabla \cdot \mathbf{V}(\mathbf{r}) = 0$ for every $\mathbf{r} \neq \mathbf{0}$.
- Explain physically** why the divergence vanishes away from the origin, even though the flow appears to “spread out” radially.
- Reconcile** this with the fact that there is a net outward flux through any sphere centered at the origin (the sphere of radius R):

$$\oint_{S_R} \mathbf{V} \cdot d\mathbf{A} \neq 0,$$

and state what this implies about the divergence at $\mathbf{r} = \mathbf{0}$ (in the distributional sense). [Hint: Use divergence theorem on the sphere of radius R to understand the paradox.]

- (d) Consider the same vector field as above and compute the closed loop line integral over the following 2 paths and note the observations. Also use the stokes theorem to compute the same and try to understand what the specialty of this particular vector field is.



4. The temperature distribution in 3D space is given by $T(x, y, z) = x + y + z$
 Constraint Surface: A thin metal sheet occupies the ellipsoid surface $g(x, y, z) = \frac{x^2}{4} + y^2 + z^2 - 1 = 0$
 Find the points on the metallic surface where it is the hottest and coldest respectively.
5. A room has a non-uniform temperature distribution given in cylindrical coordinates by

$$T(\rho, \phi, z) = 300 + 2z \quad \text{K.}$$

Inside the room, a spiral staircase of constant radius $r = 2$ m connects the ground floor at $z = 0$ to the top floor at $z = 10$ m. The staircase makes 5 complete turns while ascending. A person carries a block of ice along the staircase from the ground floor to the top floor. At the starting point, the ice is at its melting temperature 0°C . The ice melts according to the model

$$\frac{dm}{ds} = k[T(\rho, \phi, z) - 273],$$

where $k = 5 \times 10^{-4}$ kg/(m·K), m is the mass of ice, and s is the physical path length along the staircase. Assume that ice remains at 273 K throughout the motion, heat transfer depends only on the local temperature and walking speed is irrelevant. If the mass of ice remaining at the top floor must be exactly 1 kg, determine the initial mass of ice that must be carried from the ground floor.

6. Consider the region in $\mathbb{R}^3$ defined by

$$2x - y + 2z \leq 5.$$

- (a) Find the volume of the largest sphere

$$x^2 + y^2 + z^2 = r^2$$

centered at the origin that lies entirely inside this region.

- (b) Write the plane as a dot product

$$\vec{n} \cdot \vec{r} = 5, \quad \text{where } \vec{n} = (2, -1, 2), \quad \vec{r} = (x, y, z).$$

Use the Cauchy–Schwarz inequality

$$|\vec{n} \cdot \vec{r}| \leq |\vec{n}| |\vec{r}|$$

to obtain a bound on r .

You will find that Cauchy–Schwarz naturally gives a *lower bound* on $|\vec{r}|$ for points on the plane, but the problem requires an *upper bound* on r for points inside the half-space.

Explain why the inequality must be interpreted in the opposite direction in this problem, and give a geometric interpretation of the equality case.

7. A vector field is given in spherical coordinates by

$$\vec{F}(r, \theta, \phi) = \frac{\cos \phi}{r^2} \hat{r} + \frac{\sin \phi}{r^2 \sin \theta} \hat{\phi}.$$

- (a) Plot the magnitude $|\vec{F}|$ as:

- a function of ϕ for $r = 1$ and $\theta = \pi/3$,
- a function of r for $\phi = \pi/4$ and $\theta = \pi/2$.

- (b) Compute by hand and visualize them using MATLAB:

- the divergence $\nabla \cdot \vec{F}$,
- the curl $\nabla \times \vec{F}$.

- (c) Check whether the vector field $\vec{F}$ is conservative using **two independent methods**, state your final conclusion and clearly specify the region over which your conclusion is valid.

8. An Oceanographer measures the surface velocity to be

$$\vec{v}(x, y) = x^2 \hat{j}$$

Compute the curl of the velocity field at $(2, 0)$. Why is it non zero even though all the velocity vectors point in the same direction?

9. A scalar field is defined in cylindrical coordinates by

$$\Phi(\rho, \phi, z) = \rho^2 z.$$

- (a) Compute the gradient $\nabla\Phi$ in cylindrical coordinates.
(b) Evaluate the line integral

$$\int_C \nabla\Phi \cdot d\mathbf{l}$$

where C is the helix defined by

$$\rho = a, \quad \phi = t, \quad z = bt, \quad 0 \leq t \leq 2\pi.$$

- (c) Verify your result using the fundamental theorem of line integrals, and comment on the physical significance.